

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark.

```
[record 1 of 78]
  dataset : phiusiil
  feature  : URLSimilarityIndex
  accuracy : 0.9967374960991802
  precision : 1.0
  recall   : 0.992344051661008
  f1       : 0.9961573161359308
mutual_information : 0.662215786810744

[record 2 of 78]
  dataset : phiusiil
  feature  : NoOfExternalRef
  accuracy : 0.9611336491815372
  precision : 0.9394193008433658
  recall   : 0.9714399840223686
  f1       : 0.955161353668914
mutual_information : 0.5654485944505745

[record 3 of 78]
  dataset : phiusiil
  feature  : LineOfCode
  accuracy : 0.9609350619875742
  precision : 0.9653478854024556
  recall   : 0.9421476599427469
  f1       : 0.9536066844109026
mutual_information : 0.595168316462994

[record 4 of 78]
  dataset : phiusiil
  feature  : NoOfSelfRef
  accuracy : 0.948707764759284
  precision : 0.9019774870702768
  recall   : 0.9868850276279876
  f1       : 0.9425228891149542
mutual_information : 0.5293939297844878

[record 5 of 78]
  dataset : phiusiil
  feature  : NoOfImage
  accuracy : 0.9421260177593692
  precision : 0.9423430791249232
  recall   : 0.920511284202117
  f1       : 0.9312992523742172
mutual_information : 0.5494509954646463

[record 6 of 78]
  dataset : phiusiil
  feature  : NoOfJS
  accuracy : 0.9319697012681212
  precision : 0.9302610948258232
  recall   : 0.9084614872511816
  f1       : 0.9192320646682384
mutual_information : 0.504826032305074

[record 7 of 78]
  dataset : phiusiil
  feature  : NoOfCSS
  accuracy : 0.9018695565831656
  precision : 0.9678698607963744
  recall   : 0.7961520537913588
  f1       : 0.8736530664426343
mutual_information : 0.4556611168670177
```

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

```
[record 8 of 78]
  dataset : phiusiil
  feature  : HasSocialNet
  accuracy : 0.8826633379670346
  precision : 0.7863720073664825
  recall   : 0.9949404167498836
  f1       : 0.8784458943161112
  mutual_information : 0.3911324693936928

[record 9 of 78]
  dataset : phiusiil
  feature  : HasCopyrightInfo
  accuracy : 0.8674288632301626
  precision : 0.7883736484226953
  recall   : 0.9416816456960256
  f1       : 0.8582349907472014
  mutual_information : 0.3219704649717501

[record 10 of 78]
  dataset : phiusiil
  feature  : LargestLineLength
  accuracy : 0.8384635025107096
  precision : 0.9557314570507184
  recall   : 0.6510884761334133
  f1       : 0.7745307674031836
  mutual_information : 0.4585741629380953

[record 11 of 78]
  dataset : phiusiil
  feature  : HasDescription
  accuracy : 0.8328463218814718
  precision : 0.7326808380486313
  recall   : 0.956860395446375
  f1       : 0.8298978001039321
  mutual_information : 0.2806251321292377

[record 12 of 78]
  dataset : phiusiil
  feature  : NoOfOtherSpecialCharsInURL
  accuracy : 0.793838123067321
  precision : 0.9352267624607096
  recall   : 0.554623527062113
  f1       : 0.6963099168373104
  mutual_information : 0.2347809746872036

[record 13 of 78]
  dataset : phiusiil
  feature  : IsHTTPS
  accuracy : 0.791568555136316
  precision : 1.0
  recall   : 0.5108847613341322
  f1       : 0.6762723066754792
  mutual_information : 0.2348072550255246

[record 14 of 78]
  dataset : phiusiil
  feature  : DomainTitleMatchScore
  accuracy : 0.789866379188062
  precision : 0.7191205249222977
  recall   : 0.8317688569336262
  f1       : 0.7713536039512271
```

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

mutual_information : 0.1912685189780731

[record 15 of 78]

dataset : phiusiil
feature : IsResponsive
accuracy : 0.7814122386450679
precision : 0.7770372614359286
recall : 0.683043738765728
f1 : 0.7270150575730735

mutual_information : 0.1665703678319596

[record 16 of 78]

dataset : uci
feature : sslfinal_state
accuracy : 0.8735763097949886
precision : 0.90521327014218
recall : 0.8432671081677704
f1 : 0.8731428571428571

mutual_information : 0.3447768589996409

[record 17 of 78]

dataset : uci
feature : url_of_anchor
accuracy : 0.8154897494305239
precision : 0.9965870307167236
recall : 0.6445916114790287
f1 : 0.7828418230563002

mutual_information : 0.3104458656375795

[record 18 of 78]

dataset : uci
feature : having_sub_domain
accuracy : 0.6845102505694761
precision : 0.6506849315068494
recall : 0.8388520971302428
f1 : 0.7328833172613307

mutual_information : 0.0806267546036059

[record 19 of 78]

dataset : uci
feature : web_traffic
accuracy : 0.6674259681093394
precision : 0.6709129511677282
recall : 0.6975717439293598
f1 : 0.683982683982684

mutual_information : 0.058522520357888

[record 20 of 78]

dataset : uci
feature : prefix_suffix
accuracy : 0.6378132118451025
precision : 0.5875486381322957
recall : 1.0
f1 : 0.7401960784313726

mutual_information : 0.104921861669007

[record 21 of 78]

dataset : uci
feature : sfh
accuracy : 0.6150341685649203
precision : 0.58887171561051
recall : 0.8410596026490066

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

```
f1 : 0.6927272727272727
mutual_information : 0.0462648656175914

[record 22 of 78]
  dataset : uci
  feature : links_in_tags
  accuracy : 0.5968109339407744
  precision : 0.6460176991150443
  recall : 0.4834437086092715
  f1 : 0.553030303030303
mutual_information : 0.0393409056227036

[record 23 of 78]
  dataset : uci
  feature : request_url
  accuracy : 0.591116173120729
  precision : 0.6236842105263158
  recall : 0.5231788079470199
  f1 : 0.5690276110444178
mutual_information : 0.032647679112516

[record 24 of 78]
  dataset : uci
  feature : domain_registration_length
  accuracy : 0.5842824601366743
  precision : 0.6506849315068494
  recall : 0.4194260485651214
  f1 : 0.5100671140939598
mutual_information : 0.0179794658834226

[record 25 of 78]
  dataset : uci
  feature : having_ip_address
  accuracy : 0.571753986332574
  precision : 0.5955334987593052
  recall : 0.5298013245033113
  f1 : 0.5607476635514018
mutual_information : 0.0

[record 26 of 78]
  dataset : uci
  feature : links_pointing_to_page
  accuracy : 0.5637813211845103
  precision : 0.5708502024291497
  recall : 0.6225165562913907
  f1 : 0.5955649419218585
mutual_information : 0.005553214272923

[record 27 of 78]
  dataset : uci
  feature : google_index
  accuracy : 0.5318906605922551
  precision : 0.63125
  recall : 0.2229580573951435
  f1 : 0.3295269168026101
mutual_information : 0.0

[record 28 of 78]
  dataset : uci
  feature : age_of_domain
  accuracy : 0.5296127562642369
  precision : 0.5444444444444444
```

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

```

recall : 0.5408388520971302
f1 : 0.5426356589147286
mutual_information : 0.0090995515578689

```

```

[record 29 of 78]
dataset : uci
feature : abnormal_url
accuracy : 0.5296127562642369
precision : 0.5268817204301075
recall : 0.8653421633554084
f1 : 0.6549707602339181
mutual_information : 0.0129948777383075

```

```

[record 30 of 78]
dataset : uci
feature : statistical_report
accuracy : 0.5250569476082004
precision : 0.6363636363636364
recall : 0.1854304635761589
f1 : 0.2871794871794871
mutual_information : 0.0

```

```

[record 31 of 78]
dataset : mendeley_hannousse
feature : google_index
accuracy : 0.8548104956268222
precision : nan
recall : nan
f1 : 0.8611266034578918
mutual_information : 0.2994122171059308

```

```

[record 32 of 78]
dataset : mendeley_hannousse
feature : web_traffic
accuracy : 0.7877551020408163
precision : nan
recall : nan
f1 : 0.7863849765258216
mutual_information : 0.2984626354884387

```

```

[record 33 of 78]
dataset : mendeley_hannousse
feature : nb_hyperlinks
accuracy : 0.7772594752186589
precision : nan
recall : nan
f1 : 0.7983104540654699
mutual_information : 0.2336224271865306

```

```

[record 34 of 78]
dataset : mendeley_hannousse
feature : page_rank
accuracy : 0.7714285714285715
precision : nan
recall : nan
f1 : 0.7615571776155717
mutual_information : 0.2211787638339284

```

```

[record 35 of 78]
dataset : mendeley_hannousse
feature : ratio_inHyperlinks
accuracy : 0.7288629737609329

```

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

```

precision : nan
recall : nan
f1 : 0.6978557504873294
mutual_information : 0.2444913886895223

[record 36 of 78]
dataset : mendeley_hannousse
feature : nb_www
accuracy : 0.7276967930029155
precision : nan
recall : nan
f1 : 0.7426997245179063
mutual_information : 0.1107767520171292

[record 37 of 78]
dataset : mendeley_hannousse
feature : ratio_extHyperlinks
accuracy : 0.7247813411078717
precision : nan
recall : nan
f1 : 0.7114914425427873
mutual_information : 0.2402741493444735

[record 38 of 78]
dataset : mendeley_hannousse
feature : safe_anchor
accuracy : 0.7201166180758017
precision : nan
recall : nan
f1 : 0.7306397306397306
mutual_information : 0.1792403962425659

[record 39 of 78]
dataset : mendeley_hannousse
feature : longest_word_path
accuracy : 0.6973760932944606
precision : nan
recall : nan
f1 : 0.6670942912123156
mutual_information : 0.1199855314109274

[record 40 of 78]
dataset : mendeley_hannousse
feature : links_in_tags
accuracy : 0.6903790087463557
precision : nan
recall : nan
f1 : 0.7093596059113301
mutual_information : 0.1209079961488006

[record 41 of 78]
dataset : mendeley_hannousse
feature : ratio_extRedirection
accuracy : 0.6892128279883382
precision : nan
recall : nan
f1 : 0.7003934794828556
mutual_information : 0.1208953249578295

[record 42 of 78]
dataset : mendeley_hannousse
feature : domain_age

```

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

```
accuracy : 0.6880466472303207
precision : nan
recall : nan
f1 : 0.6555054732775274
mutual_information : 0.279647785006152

[record 43 of 78]
dataset : mendeley_hannousse
feature : shortest_word_host
accuracy : 0.6810495626822157
precision : nan
recall : nan
f1 : 0.6806771745475774
mutual_information : 0.0738478244125739

[record 44 of 78]
dataset : mendeley_hannousse
feature : char_repeat
accuracy : 0.6740524781341107
precision : nan
recall : nan
f1 : 0.680022896393818
mutual_information : 0.08819377638949

[record 45 of 78]
dataset : mendeley_hannousse
feature : ratio_digits_url
accuracy : 0.673469387755102
precision : nan
recall : nan
f1 : 0.6534653465346535
mutual_information : 0.1020362646838448

[record 46 of 78]
dataset : mendeley_tan
feature : PctExtHyperlinks
accuracy : 0.8886666666666667
precision : nan
recall : nan
f1 : 0.8876933422999328
mutual_information : 0.4694645544898392

[record 47 of 78]
dataset : mendeley_tan
feature : PctExtResourceUrls
accuracy : 0.7793333333333333
precision : nan
recall : nan
f1 : 0.7535368577810871
mutual_information : 0.2938791675531358

[record 48 of 78]
dataset : mendeley_tan
feature : PctExtNullSelfRedirectHyperlinksRT
accuracy : 0.7793333333333333
precision : nan
recall : nan
f1 : 0.7257663628831814
mutual_information : 0.2166061331654218

[record 49 of 78]
dataset : mendeley_tan
```

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

```
feature : PctNullSelfRedirectHyperlinks
accuracy : 0.7353333333333333
precision : nan
recall : nan
f1 : 0.7800554016620499
mutual_information : 0.2341867873312832
```

```
[record 50 of 78]
dataset : mendeley_tan
feature : ExtMetaScriptLinkRT
accuracy : 0.7306666666666667
precision : nan
recall : nan
f1 : 0.7548543689320388
mutual_information : 0.1217853502396473
```

```
[record 51 of 78]
dataset : mendeley_tan
feature : NumNumericChars
accuracy : 0.7226666666666667
precision : nan
recall : nan
f1 : 0.7409713574097135
mutual_information : 0.1245147404157263
```

```
[record 52 of 78]
dataset : mendeley_tan
feature : FrequentDomainNameMismatch
accuracy : 0.6986666666666667
precision : nan
recall : nan
f1 : 0.5830258302583026
mutual_information : 0.1318419695965218
```

```
[record 53 of 78]
dataset : mendeley_tan
feature : NumDash
accuracy : 0.6693333333333333
precision : nan
recall : nan
f1 : 0.7259668508287292
mutual_information : 0.1132495764307262
```

```
[record 54 of 78]
dataset : mendeley_tan
feature : SubmitInfoToEmail
accuracy : 0.6346666666666667
precision : nan
recall : nan
f1 : 0.7295162882527148
mutual_information : 0.077231458698451
```

```
[record 55 of 78]
dataset : mendeley_tan
feature : InsecureForms
accuracy : 0.6246666666666667
precision : nan
recall : nan
f1 : 0.7189216175736396
mutual_information : 0.0586641863473267
```

```
[record 56 of 78]
```

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

```

dataset : mendeley_tan
feature : PathLevel
accuracy : 0.6233333333333333
precision : nan
recall : nan
f1 : 0.6686217008797654
mutual_information : 0.0628910137177032

[record 57 of 78]
dataset : mendeley_tan
feature : IFrameOrFrame
accuracy : 0.62
precision : nan
recall : nan
f1 : 0.6750285062713797
mutual_information : 0.0355050630909328

[record 58 of 78]
dataset : mendeley_tan
feature : UrlLength
accuracy : 0.6173333333333333
precision : nan
recall : nan
f1 : 0.6550480769230769
mutual_information : 0.0419254325406448

[record 59 of 78]
dataset : mendeley_tan
feature : NumDots
accuracy : 0.6153333333333333
precision : nan
recall : nan
f1 : 0.5612167300380229
mutual_information : 0.0611992759683925

[record 60 of 78]
dataset : mendeley_tan
feature : PathLength
accuracy : 0.604
precision : nan
recall : nan
f1 : 0.6925465838509317
mutual_information : 0.0546984906819405

[record 61 of 78]
dataset : uci379_website_phishing
feature : SFH
accuracy : 0.8325123152709359
precision : nan
recall : nan
f1 : 0.8583333333333333
mutual_information : 0.2562112801883558

[record 62 of 78]
dataset : uci379_website_phishing
feature : SSLfinal_State
accuracy : 0.7881773399014779
precision : nan
recall : nan
f1 : 0.8138528138528138
mutual_information : 0.1673532319129793

```

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

```
[record 63 of 78]
  dataset : uci379_website_phishing
  feature  : popUpWindow
  accuracy : 0.7438423645320197
  precision : nan
  recall   : nan
  f1       : 0.7886178861788617
mutual_information : 0.1469565851250001

[record 64 of 78]
  dataset : uci379_website_phishing
  feature  : URL_Length
  accuracy : 0.6157635467980296
  precision : nan
  recall   : nan
  f1       : 0.7045454545454546
mutual_information : 0.0175734580203541

[record 65 of 78]
  dataset : uci379_website_phishing
  feature  : web_traffic
  accuracy : 0.6059113300492611
  precision : nan
  recall   : nan
  f1       : 0.6923076923076923
mutual_information : 0.0371275650453013

[record 66 of 78]
  dataset : uci379_website_phishing
  feature  : having_IP_Address
  accuracy : 0.5960591133004927
  precision : nan
  recall   : nan
  f1       : 0.7469135802469136
mutual_information : 0.0

[record 67 of 78]
  dataset : uci379_website_phishing
  feature  : Request_URL
  accuracy : 0.5911330049261084
  precision : nan
  recall   : nan
  f1       : 0.6556016597510373
mutual_information : 0.0467234326648795

[record 68 of 78]
  dataset : uci379_website_phishing
  feature  : age_of_domain
  accuracy : 0.5862068965517241
  precision : nan
  recall   : nan
  f1       : 0.6585365853658537
mutual_information : 0.0

[record 69 of 78]
  dataset : uci379_website_phishing
  feature  : URL_of_Anchor
  accuracy : 0.5763546798029556
  precision : nan
  recall   : nan
  f1       : 0.6293103448275862
mutual_information : 0.0406746833848628
```

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

[record 70 of 78]
dataset : uci379_website_phishing__suspicious_dropped
feature : SFH
accuracy : 0.8776595744680851
precision : nan
recall : nan
f1 : 0.8959276018099548
mutual_information : 0.347106157184478

[record 71 of 78]
dataset : uci379_website_phishing__suspicious_dropped
feature : SSLfinal_State
accuracy : 0.8351063829787234
precision : nan
recall : nan
f1 : 0.8544600938967136
mutual_information : 0.158212415188756

[record 72 of 78]
dataset : uci379_website_phishing__suspicious_dropped
feature : popUpWindow
accuracy : 0.75
precision : nan
recall : nan
f1 : 0.7873303167420814
mutual_information : 0.1731931852559332

[record 73 of 78]
dataset : uci379_website_phishing__suspicious_dropped
feature : URL_of_Anchor
accuracy : 0.6276595744680851
precision : nan
recall : nan
f1 : 0.6788990825688074
mutual_information : 0.0638068842395207

[record 74 of 78]
dataset : uci379_website_phishing__suspicious_dropped
feature : Request_URL
accuracy : 0.5957446808510638
precision : nan
recall : nan
f1 : 0.5064935064935064
mutual_information : 0.0496499056762813

[record 75 of 78]
dataset : uci379_website_phishing__suspicious_dropped
feature : URL_Length
accuracy : 0.5797872340425532
precision : nan
recall : nan
f1 : 0.6609442060085837
mutual_information : 0.0105902350567534

[record 76 of 78]
dataset : uci379_website_phishing__suspicious_dropped
feature : web_traffic
accuracy : 0.5691489361702128
precision : nan
recall : nan
f1 : 0.6462882096069869

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

mutual_information : 0.0622995873948881

[record 77 of 78]

dataset : uci379_website_phishing__suspicious_dropped

feature : age_of_domain

accuracy : 0.5691489361702128

precision : nan

recall : nan

f1 : 0.6301369863013698

mutual_information : 0.0527353726561756

[record 78 of 78]

dataset : uci379_website_phishing__suspicious_dropped

feature : having_IP_Address

accuracy : 0.5638297872340425

precision : nan

recall : nan

f1 : 0.7210884353741497

mutual_information : 0.0

Table T4. Progressive removal on PhiUSIL: held-out performance after greedily removing the highest-attributed feature at each iteration.

```
[record 1 of 15]
    iteration : 0
    n_features_remaining : 50
    removed_feature : URLSimilarityIndex
    accuracy : 1.0
    fl : 1.0
    mcc : 1.0
    top_feature_after : LineOfCode
    removed_so_far : 0
removed_features_so_far : nan
    top_feature_before : URLSimilarityIndex
    top_shap_before : 0.247806890352753
    second_feature : LineOfCode
    second_shap : 0.1023126702474286
    fit_seconds : 21.82
    shap_seconds : 0.01
hyperparameter_regime : retuned_per_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                    therefore understate their feature set

[record 2 of 15]
    iteration : 1
    n_features_remaining : 49
    removed_feature : LineOfCode
    accuracy : 0.9997163040086244
    fl : 0.9996671105193076
    mcc : 0.9994199473420988
    top_feature_after : IsHTTPS
    removed_so_far : 1
removed_features_so_far : URLSimilarityIndex
    top_feature_before : LineOfCode
    top_shap_before : 0.1152708303479257
    second_feature : IsHTTPS
    second_shap : 0.109668183553169
    fit_seconds : 30.77
    shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                    therefore understate their feature set

[record 3 of 15]
    iteration : 2
    n_features_remaining : 48
    removed_feature : IsHTTPS
    accuracy : 0.9995460864137988
    fl : 0.9994672704268496
    mcc : 0.9990719648501164
    top_feature_after : NoOfSelfRef
    removed_so_far : 2
removed_features_so_far : URLSimilarityIndex; LineOfCode
    top_feature_before : IsHTTPS
    top_shap_before : 0.126553705349837
    second_feature : NoOfSelfRef
    second_shap : 0.0625032237839399
    fit_seconds : 28.21
    shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                    therefore understate their feature set

[record 4 of 15]
    iteration : 3
```

Table T4. Progressive removal on PhiUSIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

```
n_features_remaining : 47
  removed_feature : NoOfSelfRef
    accuracy : 0.9981559760560584
    f1 : 0.9978342718155466
    mcc : 0.9962301516836702
  top_feature_after : NoOfExternalRef
  removed_so_far : 3
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS
top_feature_before : NoOfSelfRef
top_shap_before : 0.1063587436788972
second_feature : NoOfImage
second_shap : 0.0736897472085234
fit_seconds : 29.99
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 5 of 15]
  iteration : 4
n_features_remaining : 46
  removed_feature : NoOfExternalRef
    accuracy : 0.9978439104655452
    f1 : 0.9974661598986464
    mcc : 0.9955936724655122
  top_feature_after : NoOfImage
  removed_so_far : 4
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef
top_feature_before : NoOfExternalRef
top_shap_before : 0.0922275283717792
second_feature : NoOfImage
second_shap : 0.0850857766769547
fit_seconds : 30.4
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 6 of 15]
  iteration : 5
n_features_remaining : 45
  removed_feature : NoOfImage
    accuracy : 0.9977304320689948
    f1 : 0.997332977730364
    mcc : 0.995361320722614
  top_feature_after : HasSocialNet
  removed_so_far : 5
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef
top_feature_before : NoOfImage
top_shap_before : 0.1030192177679219
second_feature : HasSocialNet
second_shap : 0.0522758004645366
fit_seconds : 30.87
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 7 of 15]
  iteration : 6
n_features_remaining : 44
  removed_feature : HasSocialNet
```

Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

```

accuracy : 0.9973616272802064
f1 : 0.9968986560843032
mcc : 0.9946080630085278
top_feature_after : NoOfJS
removed_so_far : 6
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage
top_feature_before : HasSocialNet
top_shap_before : 0.0748647143718659
second_feature : NoOfJS
second_shap : 0.0727016331505891
fit_seconds : 30.32
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                therefore understate their feature set

[record 8 of 15]
iteration : 7
n_features_remaining : 43
removed_feature : NoOfJS
accuracy : 0.9969928224914182
f1 : 0.9964654884961652
mcc : 0.9938532601136494
top_feature_after : NoOfCSS
removed_so_far : 7
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet
top_feature_before : NoOfJS
top_shap_before : 0.0854280947188425
second_feature : NoOfOtherSpecialCharsInURL
second_shap : 0.0511721463435716
fit_seconds : 32.01
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                therefore understate their feature set

[record 9 of 15]
iteration : 8
n_features_remaining : 42
removed_feature : NoOfCSS
accuracy : 0.9965672785043548
f1 : 0.9959622251142924
mcc : 0.9929862970428124
top_feature_after : LargestLineLength
removed_so_far : 8
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS
top_feature_before : NoOfCSS
top_shap_before : 0.0897294430616568
second_feature : NoOfOtherSpecialCharsInURL
second_shap : 0.0529015924619612
fit_seconds : 27.76
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                therefore understate their feature set

[record 10 of 15]
iteration : 9
n_features_remaining : 41
removed_feature : LargestLineLength

```

Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

```

        accuracy : 0.995375755340577
        f1 : 0.9945610464146284
        mcc : 0.9905481677112188
    top_feature_after : LetterRatioInURL
    removed_so_far : 9
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS; NoOfCSS
    top_feature_before : LargestLineLength
    top_shap_before : 0.0607343714721147
    second_feature : LetterRatioInURL
    second_shap : 0.0535198590666558
    fit_seconds : 23.92
    shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                  therefore understate their feature set

[record 11 of 15]
    iteration : 10
    n_features_remaining : 40
    removed_feature : LetterRatioInURL
        accuracy : 0.9945814065647252
        f1 : 0.9936237689868136
        mcc : 0.9889254016453218
    top_feature_after : NoOfOtherSpecialCharsInURL
    removed_so_far : 10
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength
    top_feature_before : LetterRatioInURL
    top_shap_before : 0.0583808986068427
    second_feature : NoOfOtherSpecialCharsInURL
    second_shap : 0.054602624538909
    fit_seconds : 23.86
    shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                  therefore understate their feature set

[record 12 of 15]
    iteration : 11
    n_features_remaining : 39
    removed_feature : NoOfOtherSpecialCharsInURL
        accuracy : 0.9927657522199212
        f1 : 0.9914826814522864
        mcc : 0.9852132882490036
    top_feature_after : NoOfLettersInURL
    removed_so_far : 11
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL
    top_feature_before : NoOfOtherSpecialCharsInURL
    top_shap_before : 0.0654809384262721
    second_feature : HasCopyrightInfo
    second_shap : 0.0521447968456687
    fit_seconds : 24.34
    shap_seconds : 1718.39
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                  therefore understate their feature set

[record 13 of 15]
    iteration : 12
    n_features_remaining : 38

```

Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

```

    removed_feature : NoOfLettersInURL
    accuracy : 0.9925671650259582
    f1 : 0.991250918319642
    mcc : 0.9848054869947998
    top_feature_after : HasCopyrightInfo
    removed_so_far : 12
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
    HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL;
    NoOfOtherSpecialCharsInURL
    top_feature_before : NoOfLettersInURL
    top_shap_before : 0.0557818683421087
    second_feature : HasCopyrightInfo
    second_shap : 0.0509389295780166
    fit_seconds : 31.67
    shap_seconds : 1597.42
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
    therefore understate their feature set

[record 14 of 15]
    iteration : 13
    n_features_remaining : 37
    removed_feature : HasCopyrightInfo
    accuracy : 0.991801185849244
    f1 : 0.990350906480585
    mcc : 0.9832372088397836
    top_feature_after : HasDescription
    removed_so_far : 13
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
    HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL;
    NoOfOtherSpecialCharsInURL; NoOfLettersInURL
    top_feature_before : HasCopyrightInfo
    top_shap_before : 0.0573962341788812
    second_feature : HasDescription
    second_shap : 0.0535061880736899
    fit_seconds : 24.72
    shap_seconds : 1571.49
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
    therefore understate their feature set

[record 15 of 15]
    iteration : 14
    n_features_remaining : 36
    removed_feature : nan
    accuracy : 0.989191182728588
    f1 : 0.9873055009495884
    mcc : 0.9778960075454448
    top_feature_after : nan
    removed_so_far : 14
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
    HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL;
    NoOfOtherSpecialCharsInURL; NoOfLettersInURL; HasCopyrightInfo
    top_feature_before : HasDescription
    top_shap_before : 0.0594322166570935
    second_feature : DomainTitleMatchScore
    second_shap : 0.0528102098854716
    fit_seconds : 33.0
    shap_seconds : 2022.07
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
    therefore understate their feature set

```

Table T4. Progressive removal on PhiUSIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

Table T6. Zero-shot transfer between benchmarks on the shared nine-concept schema, five seeds per direction.

```
[record 1 of 4]
    direction : phiusiil->phiusiil
      train   : phiusiil
      test    : phiusiil
in_distribution : True
  n_seeds    : 5
  n_test     : 35249
majority_class_accuracy : 0.5738602513546484
  accuracy_mean : 0.9019262957814408
  accuracy_std  : 0.0006598600442346
  accuracy_ci_lower : 0.9013929473176544
  accuracy_ci_upper : 0.9024369485659168
  precision_mean : 0.9056444339863938
  precision_std  : 0.0016384277650987
  precision_ci_lower : 0.9043577558234336
  precision_ci_upper : 0.9068350358020898
  recall_mean : 0.859396844417815
  recall_std  : 0.0017820063663303
  recall_ci_lower : 0.8580254310631782
  recall_ci_upper : 0.8607695892417283
  f1_mean : 0.8819126663106234
  f1_std  : 0.0008118634767744
  f1_ci_lower : 0.8813078511702008
  f1_ci_upper : 0.8825885014345717
  mcc_mean : 0.7989546295294948
  mcc_std  : 0.0013595192955533
  mcc_ci_lower : 0.7978481610074346
  mcc_ci_upper : 0.80001235371225
  roc_auc_mean : 0.9669932143117757
  roc_auc_std  : 0.0003512505464175
  roc_auc_ci_lower : 0.966729356897838
  roc_auc_ci_upper : 0.9672570717257128
accuracy_above_majority : 0.3280660444267923
```

```
[record 2 of 4]
    direction : phiusiil->uci
      train   : phiusiil
      test    : uci
in_distribution : False
  n_seeds    : 5
  n_test     : 1659
majority_class_accuracy : 0.5569620253164558
  accuracy_mean : 0.5265822784810126
  accuracy_std  : 0.0051782354659227
  accuracy_ci_lower : 0.5226039783001808
  accuracy_ci_upper : 0.531045810729355
  precision_mean : 0.4831216822174379
  precision_std  : 0.0027620802310054
  precision_ci_lower : 0.4810021693342975
  precision_ci_upper : 0.485517997823656
  recall_mean : 0.9804081632653062
  recall_std  : 0.003547865254532
  recall_ci_lower : 0.9779523809523808
  recall_ci_upper : 0.9834013605442176
  f1_mean : 0.6472720805963317
  f1_std  : 0.0022272565970481
  f1_ci_lower : 0.6455072676925744
  f1_ci_upper : 0.6490009463178119
  mcc_mean : 0.240783149229503
  mcc_std  : 0.0086634870694218
  mcc_ci_lower : 0.2337392084360741
  mcc_ci_upper : 0.2473856120508793
```

Table T6. Zero-shot transfer between benchmarks on the shared nine-concept schema, five seeds per direction. (continued)

```

roc_auc_mean : 0.8401956886650763
roc_auc_std : 0.0256361857735798
roc_auc_ci_lower : 0.818209058515181
roc_auc_ci_upper : 0.8555607091321376
accuracy_above_majority : -0.0303797468354432

```

[record 3 of 4]

```

direction : uci->uci
train : uci
test : uci
in_distribution : True
n_seeds : 5
n_test : 1659
majority_class_accuracy : 0.5569620253164558
accuracy_mean : 0.9205545509342976
accuracy_std : 0.0055146295414021
accuracy_ci_lower : 0.916576250753466
accuracy_ci_upper : 0.9250150693188668
precision_mean : 0.9284360862153792
precision_std : 0.0063231068912138
precision_ci_lower : 0.9232132415948936
precision_ci_upper : 0.933204161703942
recall_mean : 0.8892517006802722
recall_std : 0.0105650441677916
recall_ci_lower : 0.8802721088435372
recall_ci_upper : 0.8965986394557823
f1_mean : 0.9083910184922048
f1_std : 0.0066509988848052
f1_ci_lower : 0.9034051502405068
f1_ci_upper : 0.913758538162139
mcc_mean : 0.838964368316882
mcc_std : 0.0111231979990243
mcc_ci_lower : 0.8309454686098032
mcc_ci_upper : 0.8479609433037714
roc_auc_mean : 0.9740947374620844
roc_auc_std : 0.0022770168938744
roc_auc_ci_lower : 0.9724295285213652
roc_auc_ci_upper : 0.975889312218394
accuracy_above_majority : 0.3635925256178419

```

[record 4 of 4]

```

direction : uci->phiusiil
train : uci
test : phiusiil
in_distribution : False
n_seeds : 5
n_test : 35249
majority_class_accuracy : 0.5738602513546484
accuracy_mean : 0.6402053958977559
accuracy_std : 0.0113818428681579
accuracy_ci_lower : 0.6300434054866805
accuracy_ci_upper : 0.6468155124968085
precision_mean : 0.859368831430721
precision_std : 0.0165619140936406
precision_ci_lower : 0.8468175896656912
precision_ci_upper : 0.8729784745467195
recall_mean : 0.1859130550562545
recall_std : 0.0298115941240398
recall_ci_lower : 0.1598828307036815
recall_ci_upper : 0.2037014845882431
f1_mean : 0.3048298427050162
f1_std : 0.0420244690085735

```

Table T6. Zero-shot transfer between benchmarks on the shared nine-concept schema, five seeds per direction. (continued)

```
f1_ci_lower : 0.2678617797510827
f1_ci_upper : 0.3293774311645302
  mcc_mean : 0.2787652457513706
  mcc_std : 0.0297141582331254
mcc_ci_lower : 0.2521239485140789
mcc_ci_upper : 0.2961691812424171
  roc_auc_mean : 0.7144176424049767
  roc_auc_std : 0.0161339241007508
roc_auc_ci_lower : 0.7040834358494903
roc_auc_ci_upper : 0.7283761239883436
accuracy_above_majority : 0.0663451445431075
```

Table T7. Within-dataset ceiling: Kendall's tau between attribution rankings from independent models on disjoint halves of the same data.

[record 1 of 4]

```
config : phiusiil_full
dataset : phiusiil
n_seeds : 5
n_features : 50
top1_agreement_rate : 1.0
n_features_sign_top10_mean : 9.2
kendall_tau_mean : 0.9190204081632654
kendall_tau_std : 0.0175310755207132
kendall_tau_ci_lower : 0.9053061224489796
kendall_tau_ci_upper : 0.9327346938775511
jaccard_5_mean : 0.9333333333333332
jaccard_5_std : 0.149071198499986
jaccard_5_ci_lower : 0.7999999999999999
jaccard_5_ci_upper : 1.0
jaccard_10_mean : 0.8606060606060606
jaccard_10_std : 0.1414862730900455
jaccard_10_ci_lower : 0.7575757575757576
jaccard_10_ci_upper : 0.9636363636363636
jaccard_15_mean : 0.8529411764705882
jaccard_15_std : 0.0493250289154365
jaccard_15_ci_lower : 0.8088235294117647
jaccard_15_ci_upper : 0.875
sign_agreement_top5_mean : 34.0
sign_agreement_top5_std : 24.08318915758459
sign_agreement_top5_ci_lower : 12.0
sign_agreement_top5_ci_upper : 52.0
sign_agreement_top10_mean : 41.22222222222221
sign_agreement_top10_std : 19.02889291136241
sign_agreement_top10_ci_lower : 26.866666666666667
sign_agreement_top10_ci_upper : 55.58333333333331
sign_agreement_top15_mean : 41.86813186813187
sign_agreement_top15_std : 13.251899814357632
sign_agreement_top15_ci_lower : 33.73626373626374
sign_agreement_top15_ci_upper : 54.28571428571429
sign_agreement_weighted_mean : 26.24985992335248
sign_agreement_weighted_std : 30.18856396281425
sign_agreement_weighted_ci_lower : 5.249993965166468
sign_agreement_weighted_ci_upper : 52.81873711378464
sign_agreement_unfiltered_mean : 43.6
sign_agreement_unfiltered_std : 11.260550608207398
sign_agreement_unfiltered_ci_lower : 36.0
sign_agreement_unfiltered_ci_upper : 53.2
```

[record 2 of 4]

```
config : phiusiil_leakfree
dataset : phiusiil
n_seeds : 5
n_features : 36
top1_agreement_rate : 0.8
n_features_sign_top10_mean : 9.6
kendall_tau_mean : 0.9307936507936508
kendall_tau_std : 0.0098872453983475
kendall_tau_ci_lower : 0.9225396825396828
kendall_tau_ci_upper : 0.9371428571428572
jaccard_5_mean : 0.8666666666666666
jaccard_5_std : 0.1825741858350553
jaccard_5_ci_lower : 0.7333333333333332
jaccard_5_ci_upper : 1.0
jaccard_10_mean : 0.9272727272727274
jaccard_10_std : 0.0995859195463938
jaccard_10_ci_lower : 0.8545454545454545
```

Table T7. Within-dataset ceiling: Kendall's tau between attribution rankings from independent models on disjoint halves of the same data. (continued)

```
jaccard_10_ci_upper : 1.0
jaccard_15_mean : 0.9
jaccard_15_std : 0.0559016994374947
jaccard_15_ci_lower : 0.875
jaccard_15_ci_upper : 0.95
sign_agreement_top5_mean : 62.0
sign_agreement_top5_std : 30.331501776206203
sign_agreement_top5_ci_lower : 38.0
sign_agreement_top5_ci_upper : 84.04999999999995
sign_agreement_top10_mean : 51.333333333333336
sign_agreement_top10_std : 21.17534088812728
sign_agreement_top10_ci_lower : 35.33333333333333
sign_agreement_top10_ci_upper : 68.0
sign_agreement_top15_mean : 44.0
sign_agreement_top15_std : 16.79407086821211
sign_agreement_top15_ci_lower : 32.0
sign_agreement_top15_ci_upper : 55.71428571428571
sign_agreement_weighted_mean : 47.13274094309819
sign_agreement_weighted_std : 19.149237951466915
sign_agreement_weighted_ci_lower : 33.085354004943284
sign_agreement_weighted_ci_upper : 61.737679147962425
sign_agreement_unfiltered_mean : 48.88888888888889
sign_agreement_unfiltered_std : 11.719457283182768
sign_agreement_unfiltered_ci_lower : 40.55555555555556
sign_agreement_unfiltered_ci_upper : 58.88888888888888
```

[record 3 of 4]

```
config : uci_dedup
dataset : uci
n_seeds : 5
n_features : 30
top1_agreement_rate : 0.4
n_features_sign_top10_mean : 9.0
kendall_tau_mean : 0.7682758620689655
kendall_tau_std : 0.0115401382970217
kendall_tau_ci_lower : 0.759080459770115
kendall_tau_ci_upper : 0.7765747126436782
jaccard_5_mean : 0.6666666666666666
jaccard_5_std : 0.0
jaccard_5_ci_lower : 0.6666666666666666
jaccard_5_ci_upper : 0.6666666666666666
jaccard_10_mean : 0.8181818181818181
jaccard_10_std : 1.2412670766236366e-16
jaccard_10_ci_lower : 0.8181818181818181
jaccard_10_ci_upper : 0.8181818181818181
jaccard_15_mean : 0.7892156862745098
jaccard_15_std : 0.0879454662647507
jaccard_15_ci_lower : 0.7254901960784312
jaccard_15_ci_upper : 0.8529411764705882
sign_agreement_top5_mean : 50.0
sign_agreement_top5_std : 17.67766952966369
sign_agreement_top5_ci_lower : 35.0
sign_agreement_top5_ci_upper : 65.0
sign_agreement_top10_mean : 44.44444444444444
sign_agreement_top10_std : 17.568209223157663
sign_agreement_top10_ci_lower : 31.111111111111107
sign_agreement_top10_ci_upper : 57.77777777777777
sign_agreement_top15_mean : 51.684981684981686
sign_agreement_top15_std : 12.909424815646569
sign_agreement_top15_ci_lower : 41.75824175824176
sign_agreement_top15_ci_upper : 61.61172161172161
sign_agreement_weighted_mean : 55.69565427371174
```

Table T7. Within-dataset ceiling: Kendall's tau between attribution rankings from independent models on disjoint halves of the same data. (continued)

```
sign_agreement_weighted_std : 13.867915059366505
sign_agreement_weighted_ci_lower : 45.54650394227103
sign_agreement_weighted_ci_upper : 67.4256585842625
sign_agreement_unfiltered_mean : 50.66666666666667
sign_agreement_unfiltered_std : 4.944132324730441
sign_agreement_unfiltered_ci_lower : 46.66666666666667
sign_agreement_unfiltered_ci_upper : 54.0
```

[record 4 of 4]

```
config : uci_full
dataset : uci
n_seeds : 5
n_features : 30
top1_agreement_rate : 0.6
n_features_sign_top10_mean : 9.6
kendall_tau_mean : 0.824367816091954
kendall_tau_std : 0.0185053901586189
kendall_tau_ci_lower : 0.8096321839080459
kendall_tau_ci_upper : 0.83816091954023
jaccard_5_mean : 0.8
jaccard_5_std : 0.1825741858350553
jaccard_5_ci_lower : 0.6666666666666666
jaccard_5_ci_upper : 0.9333333333333332
jaccard_10_mean : 0.9272727272727274
jaccard_10_std : 0.0995859195463938
jaccard_10_ci_lower : 0.8545454545454547
jaccard_10_ci_upper : 1.0
jaccard_15_mean : 0.9
jaccard_15_std : 0.0559016994374947
jaccard_15_ci_lower : 0.875
jaccard_15_ci_upper : 0.95
sign_agreement_top5_mean : 45.0
sign_agreement_top5_std : 13.228756555322953
sign_agreement_top5_ci_lower : 33.0
sign_agreement_top5_ci_upper : 54.0
sign_agreement_top10_mean : 44.0
sign_agreement_top10_std : 15.666272650679616
sign_agreement_top10_ci_lower : 32.0
sign_agreement_top10_ci_upper : 56.0
sign_agreement_top15_mean : 46.38095238095239
sign_agreement_top15_std : 11.256643289213246
sign_agreement_top15_ci_lower : 37.14285714285714
sign_agreement_top15_ci_upper : 54.095238095238095
sign_agreement_weighted_mean : 47.14090504304773
sign_agreement_weighted_std : 24.162626591280908
sign_agreement_weighted_ci_lower : 26.45745296751132
sign_agreement_weighted_ci_upper : 64.92344984195503
sign_agreement_unfiltered_mean : 46.0
sign_agreement_unfiltered_std : 4.346134936801765
sign_agreement_unfiltered_ci_lower : 42.666666666666666
sign_agreement_unfiltered_ci_upper : 49.33333333333333
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement.

```
[record 1 of 69]
  concept : ip_in_domain
  status : mapped
  uci_feature : having_ip_address
  phiusiil_source : IsDomainIP
  rationale : Both encode 'the host is a bare IP address rather than a domain name'. Direct binary
              correspondence.
  caveat : nan
  rho_phiusiil : -0.0605489999124573
  rho_uci : -0.0941600949562038
direction_agrees : True
polarity_status : agrees

[record 2 of 69]
  concept : url_length
  status : mapped
  uci_feature : url_length
  phiusiil_source : URLLength
  rationale : Same quantity (characters in the URL); PhiUSIIL stores it raw, UCI stores it binned. Discretised
              with UCI's published 54/75 thresholds.
  caveat : nan
  rho_phiusiil : -0.3539937654486287
  rho_uci : -0.0533315794339325
direction_agrees : True
polarity_status : agrees

[record 3 of 69]
  concept : subdomain_depth
  status : mapped
  uci_feature : having_sub_domain
  phiusiil_source : NoOfSubDomain
  rationale : Both encode subdomain nesting depth. UCI counts dots after stripping www and ccTLD; PhiUSIIL counts
              subdomains directly, so the counts align up to the stripping convention.
  caveat : UCI's dot-count and PhiUSIIL's subdomain count can differ by one for hosts with a ccTLD (e.g.
              .co.uk).
  rho_phiusiil : -0.1141491829083289
  rho_uci : -0.3046922676521399
direction_agrees : True
polarity_status : agrees

[record 4 of 69]
  concept : ssl_state
  status : mapped
  uci_feature : sslfinal_state
  phiusiil_source : IsHTTPS
  rationale : Closest available correspondence to UCI's SSL assessment: whether the page is served over HTTPS at
              all, which is the dominant component of sslfinal_state.
  caveat : IMPERFECT. UCI's sslfinal_state additionally requires a trusted issuer and a certificate at least a
              year old, and has a 'suspicious' state for https-with-untrusted-issuer. PhiUSIIL records only the
              scheme, so the suspicious state is unreachable from the PhiUSIIL side.
  rho_phiusiil : -0.6124154229775562
  rho_uci : -0.735813975062039
direction_agrees : True
polarity_status : agrees

[record 5 of 69]
  concept : iframe
  status : mapped
  uci_feature : iframe
  phiusiil_source : NoOfiFrame
  rationale : Both encode iframe usage on the page.
  caveat : UCI targets specifically INVISIBLE iframes; PhiUSIIL counts all iframes. Note also that UCI's
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

        iframe feature carries no marginal signal at all (phishing rate 0.44 at both of its values), so
        this concept contributes nothing on the UCI side regardless of the mapping quality.
    rho_phiusiil : 0.521331928092635
    rho_uci : 0.0033935238644643
    direction_agrees : False
    polarity_status : NO_SIGNAL_UCI

[record 6 of 69]
    concept : popup
    status : mapped
    uci_feature : popupwindow
    phiusiil_source : NoOfPopup
    rationale : Both encode popup-window usage.
    caveat : UCI targets popups containing a text field; PhiUSIIL counts all popups. As with iframe, UCI's
        popupwindow has phishing rate 0.44 at both values - zero marginal signal on the UCI side.
    rho_phiusiil : 0.1925460026806499
    rho_uci : -8.588679256217027e-05
    direction_agrees : False
    polarity_status : NO_SIGNAL_UCI

[record 7 of 69]
    concept : url_redirect
    status : mapped
    uci_feature : redirect
    phiusiil_source : NoOfURLRedirect
    rationale : Both encode redirection on the way to the page. UCI's `redirect` takes only {0, 1} in the delivered
        data, so the PhiUSIIL side is mapped onto that same {0, 1} domain rather than the usual ternary
        one.
    caveat : UCI's published rule describes a 3-way split on redirect count, but the delivered column is binary;
        the mapping follows the DATA, not the paper.
    rho_phiusiil : 0.0480143528227923
    rho_uci : 0.0201134616575737
    direction_agrees : True
    polarity_status : agrees

[record 8 of 69]
    concept : form_handler
    status : mapped
    uci_feature : sfh
    phiusiil_source : HasExternalFormSubmit
    rationale : Both encode whether a form posts to a different domain than the page - UCI's Server Form Handler
        check and PhiUSIIL's external-form-submit flag.
    caveat : UCI's 'suspicious' state (blank/about:blank handler) has no PhiUSIIL counterpart and is unreachable
        from the PhiUSIIL side.
    rho_phiusiil : 0.1673089014798561
    rho_uci : -0.2191900653063038
    direction_agrees : False
    polarity_status : INVERTED

[record 9 of 69]
    concept : anchor_external_ratio
    status : mapped
    uci_feature : url_of_anchor
    phiusiil_source : NoOfExternalRef + NoOfSelfRef + NoOfEmptyRef
    rationale : UCI's url_of_anchor is the share of <a> hrefs pointing off-domain. PhiUSIIL provides the underlying
        counts (external / self / empty references), so the ratio is reconstructed directly and binned with
        UCI's 31%/67% thresholds.
    caveat : nan
    rho_phiusiil : 0.0872798521005418
    rho_uci : -0.7012072591666815
    direction_agrees : False
    polarity_status : INVERTED

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 10 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : shortening_service
  phiusiil_source : nan
  rationale : URL-shortener detection; PhiUSIIL has no shortener feature.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 11 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : having_at_symbol
  phiusiil_source : nan
  rationale : presence of '@' in the URL; PhiUSIIL counts only aggregate special characters, with no per-
               character breakdown.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 12 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : double_slash_redirecting
  phiusiil_source : nan
  rationale : '//' position in the path; no PhiUSIIL equivalent.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 13 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : prefix_suffix
  phiusiil_source : nan
  rationale : hyphen in the domain; PhiUSIIL has no hyphen feature.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 14 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : domain_registration_length
  phiusiil_source : nan
  rationale : WHOIS registration length; PhiUSIIL has no WHOIS data.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 15 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : favicon
  phiusiil_source : nan
  rationale : favicon loaded from an EXTERNAL domain. PhiUSIIL's HasFavicon records presence only, which is a
              different concept - mapping them would force a match between 'has a favicon' and 'favicon is
              foreign'.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 16 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : port
  phiusiil_source : nan
  rationale : non-standard port; no PhiUSIIL equivalent.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 17 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : https_token
  phiusiil_source : nan
  rationale : the literal token 'https' inside the domain name (a deception pattern), NOT whether HTTPS is used.
              PhiUSIIL's IsHTTPS is the latter and is already mapped to sslfinal_state.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 18 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : request_url
  phiusiil_source : nan
  rationale : share of loaded OBJECTS (images/scripts) from an external domain. PhiUSIIL has
              NoOfImage/NoOfCSS/NoOfJS but no internal/external split for them, so the ratio cannot be
              reconstructed.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 19 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : links_in_tags
  phiusiil_source : nan
  rationale : share of links inside meta/script/link tags; PhiUSIIL has no tag-scoped link counts.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

direction_agrees : nan
polarity_status : nan

[record 20 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : submitting_to_email
  phiusiil_source : nan
  rationale : mailto: form action; no PhiUSIIL equivalent.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 21 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : abnormal_url
  phiusiil_source : nan
  rationale : WHOIS hostname consistency; no PhiUSIIL WHOIS data.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 22 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : on_mouseover
  phiusiil_source : nan
  rationale : JavaScript status-bar manipulation; PhiUSIIL records no JS behaviour.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 23 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : rightclick
  phiusiil_source : nan
  rationale : right-click disabling; PhiUSIIL records no JS behaviour.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 24 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : age_of_domain
  phiusiil_source : nan
  rationale : WHOIS domain age; no PhiUSIIL WHOIS data.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 25 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : dnsrecord
  phiusiil_source : nan
  rationale : DNS record existence; no PhiUSIIL DNS data.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 26 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : web_traffic
  phiusiil_source : nan
  rationale : Alexa traffic rank; external reputation, absent from PhiUSIIL.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 27 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : page_rank
  phiusiil_source : nan
  rationale : PageRank; external reputation, absent from PhiUSIIL.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 28 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : google_index
  phiusiil_source : nan
  rationale : Google index membership; external reputation, absent from PhiUSIIL.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 29 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : links_pointing_to_page
  phiusiil_source : nan
  rationale : INBOUND link count. PhiUSIIL's NoOfExternalRef is OUTBOUND; mapping them would invert the direction
    of the relationship.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 30 of 69]
  concept : nan
  status : unmapped_uci
  uci_feature : statistical_report
  phiusiil_source : nan
  rationale : membership of a published blacklist; external reputation.
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 31 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : DomainLength
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 32 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : URLSimilarityIndex
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 33 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : CharContinuationRate
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 34 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : TLDLegitimateProb
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 35 of 69]
  concept : nan
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : URLCharProb
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 36 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : TLDLength
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 37 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : HasObfuscation
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 38 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : NoOf0bfuscatedChar
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 39 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : ObfuscationRatio
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 40 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

phiusiil_source : NoOfLettersInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 41 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
phiusiil_source : LetterRatioInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 42 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
phiusiil_source : NoOfDegitsInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 43 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
phiusiil_source : DeditRatioInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 44 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
phiusiil_source : NoOfEqualsInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 45 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
phiusiil_source : NoOfQMarkInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 46 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : NoOfAmpersandInURL
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 47 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : NoOfOtherSpecialCharsInURL
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 48 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : SpacialCharRatioInURL
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 49 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : LineOfCode
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 50 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : LargestLineLength
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 51 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : HasTitle
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 52 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : DomainTitleMatchScore
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 53 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : URLTitleMatchScore
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 54 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : HasFavicon
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 55 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : Robots
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

polarity_status : nan

[record 56 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : IsResponsive
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 57 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : NoOfSelfRedirect
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 58 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : HasDescription
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 59 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : HasSocialNet
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 60 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : HasSubmitButton
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 61 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : HasHiddenFields
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 62 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : HasPasswordField
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 63 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : Bank
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 64 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : Pay
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 65 of 69]
  concept : nan
  status : unmapped_phiusiil
  uci_feature : nan
  phiusiil_source : Crypto
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat : nan
  rho_phiusiil : nan
  rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 66 of 69]
  concept : nan
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : HasCopyrightInfo
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 67 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : NoOfImage
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 68 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : NoOfCSS
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

[record 69 of 69]
    concept : nan
    status : unmapped_phiusiil
    uci_feature : nan
    phiusiil_source : NoOfJS
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat : nan
    rho_phiusiil : nan
    rho_uci : nan
direction_agrees : nan
polarity_status : nan

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load.

[record 1 of 24]

```

dataset : phiusiil
iteration : 0
n_features : 50
accuracy : 1.0
fit_seconds : 20.46
shap_seconds : 284.19
mean_tree_depth : 13.864
total_nodes : 104862
window_start : 10:34:34
window_end : 10:38:27
cache_replay : False
source_log : a13.log
run_date : 2026-09-11
window_reconstructed : True
competing_jobs : nan
contention_evidence : remeasured_uncontended
sampled_cpu_percent_mean : nan
sampled_cpu_percent_max : nan
sampled_n_python_procs_max : nan
sampled_job_tags : nan
n_load_samples : nan
contended : False
cost_analysis_eligible : True
remeasured_clean : True
shap_seconds_original_contended : 179.9
f1 : nan
mcc : nan
removed_feature : nan
top_feature : nan
hyperparameter_regime : nan

```

[record 2 of 24]

```

dataset : phiusiil
iteration : 1
n_features : 49
accuracy : 0.999716
fit_seconds : 31.2
shap_seconds : 683.21
mean_tree_depth : 19.158
total_nodes : 281270
window_start : 10:38:27
window_end : 10:46:39
cache_replay : False
source_log : a13.log
run_date : 2026-09-11
window_reconstructed : nan
competing_jobs : nan
contention_evidence : remeasured_uncontended
sampled_cpu_percent_mean : nan
sampled_cpu_percent_max : nan
sampled_n_python_procs_max : nan
sampled_job_tags : nan
n_load_samples : nan
contended : False
cost_analysis_eligible : True
remeasured_clean : True
shap_seconds_original_contended : 463.5
f1 : nan
mcc : nan
removed_feature : nan
top_feature : nan

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

hyperparameter_regime : nan

[record 3 of 24]
    dataset : phiusiil
    iteration : 2
    n_features : 48
    accuracy : 0.999546
    fit_seconds : 27.4
    shap_seconds : 788.7
    mean_tree_depth : 20.152
    total_nodes : 367168
    window_start : 10:46:39
    window_end : 10:56:32
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : nan
    contention_evidence : remeasured_uncontended
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
    sampled_job_tags : nan
    n_load_samples : nan
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : True
shap_seconds_original_contended : 564.3
    f1 : nan
    mcc : nan
    removed_feature : nan
    top_feature : nan
    hyperparameter_regime : nan

[record 4 of 24]
    dataset : phiusiil
    iteration : 3
    n_features : 47
    accuracy : 0.998156
    fit_seconds : 28.77
    shap_seconds : 978.95
    mean_tree_depth : 23.722
    total_nodes : 544400
    window_start : 10:56:32
    window_end : 11:08:45
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : nan
    contention_evidence : remeasured_uncontended
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
    sampled_job_tags : nan
    n_load_samples : nan
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : True
shap_seconds_original_contended : 705.8
    f1 : nan
    mcc : nan

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

removed_feature : nan
top_feature : nan
hyperparameter_regime : nan

[record 5 of 24]
dataset : phiusiil
iteration : 4
n_features : 46
accuracy : 0.997844
fit_seconds : 27.9
shap_seconds : 582.7
mean_tree_depth : 25.126
total_nodes : 628132
window_start : 11:08:45
window_end : 11:18:56
cache_replay : False
source_log : a13.log
run_date : 2026-09-11
window_reconstructed : nan
competing_jobs : nan
contention_evidence : none
sampled_cpu_percent_mean : nan
sampled_cpu_percent_max : nan
sampled_n_python_procs_max : nan
sampled_job_tags : nan
n_load_samples : nan
contended : False
cost_analysis_eligible : True
remeasured_clean : False
shap_seconds_original_contended : nan
f1 : nan
mcc : nan
removed_feature : nan
top_feature : nan
hyperparameter_regime : nan

```

```

[record 6 of 24]
dataset : phiusiil
iteration : 5
n_features : 45
accuracy : 0.99773
fit_seconds : 25.2
shap_seconds : 851.9
mean_tree_depth : 26.246
total_nodes : 704950
window_start : 11:18:56
window_end : 11:33:34
cache_replay : False
source_log : a13.log
run_date : 2026-09-11
window_reconstructed : nan
competing_jobs : nan
contention_evidence : none
sampled_cpu_percent_mean : nan
sampled_cpu_percent_max : nan
sampled_n_python_procs_max : nan
sampled_job_tags : nan
n_load_samples : nan
contended : False
cost_analysis_eligible : True
remeasured_clean : False
shap_seconds_original_contended : nan

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

            f1 : nan
            mcc : nan
removed_feature : nan
            top_feature : nan
hyperparameter_regime : nan

[record 7 of 24]
    dataset : phiusiil
    iteration : 6
    n_features : 44
    accuracy : 0.997362
    fit_seconds : 25.4
    shap_seconds : 752.1
    mean_tree_depth : 28.38
    total_nodes : 838026
    window_start : 11:33:34
    window_end : 11:46:32
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : nan
    contention_evidence : none
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
    sampled_job_tags : nan
    n_load_samples : nan
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended : nan
    f1 : nan
    mcc : nan
    removed_feature : nan
    top_feature : nan
    hyperparameter_regime : nan

```

```

[record 8 of 24]
    dataset : phiusiil
    iteration : 7
    n_features : 43
    accuracy : 0.996993
    fit_seconds : 32.3
    shap_seconds : 700.5
    mean_tree_depth : 28.88
    total_nodes : 952992
    window_start : 11:46:32
    window_end : 11:58:46
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : nan
    contention_evidence : none
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
    sampled_job_tags : nan
    n_load_samples : nan
    contended : False
    cost_analysis_eligible : True

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

remeasured_clean : False
shap_seconds_original_contended : nan
    f1 : nan
    mcc : nan
    removed_feature : nan
    top_feature : nan
hyperparameter_regime : nan

[record 9 of 24]
    dataset : phiusiil
    iteration : 8
    n_features : 42
    accuracy : 0.996567
    fit_seconds : 24.1
    shap_seconds : 717.8
    mean_tree_depth : 31.152
    total_nodes : 1075554
    window_start : 11:58:46
    window_end : 12:11:08
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : nan
    contention_evidence : none
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
    sampled_job_tags : nan
    n_load_samples : nan
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended : nan
    f1 : nan
    mcc : nan
    removed_feature : nan
    top_feature : nan
hyperparameter_regime : nan

```

```

[record 10 of 24]
    dataset : phiusiil
    iteration : 9
    n_features : 41
    accuracy : 0.995376
    fit_seconds : 29.1
    shap_seconds : 1220.1
    mean_tree_depth : 33.43
    total_nodes : 1291626
    window_start : 12:11:08
    window_end : 12:31:58
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : nan
    contention_evidence : none
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
    sampled_job_tags : nan
    n_load_samples : nan

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

        contended : False
cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended : nan
    f1 : nan
    mcc : nan
    removed_feature : nan
    top_feature : nan
hyperparameter_regime : nan

```

[record 11 of 24]

```

    dataset : phiusiil
    iteration : 10
    n_features : 40
    accuracy : 0.994581
    fit_seconds : 25.4
    shap_seconds : 1264.2
    mean_tree_depth : 36.51
    total_nodes : 1496914
    window_start : 12:31:58
    window_end : 12:53:28
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : nan
    contention_evidence : none
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
    sampled_job_tags : nan
    n_load_samples : nan
    contended : False
cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended : nan
    f1 : nan
    mcc : nan
    removed_feature : nan
    top_feature : nan
hyperparameter_regime : nan

```

[record 12 of 24]

```

    dataset : phiusiil
    iteration : 11
    n_features : 39
    accuracy : 0.992766
    fit_seconds : 28.83
    shap_seconds : 1860.9
    mean_tree_depth : 39.466
    total_nodes : 1718032
    window_start : 14:59:46
    window_end : 15:28:50
    cache_replay : False
    source_log : a13_resume.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : nan
    contention_evidence : remeasured_uncontended
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

    sampled_job_tags : nan
    n_load_samples : nan
      contended : False
    cost_analysis_eligible : True
      remeasured_clean : True
    shap_seconds_original_contended : 1718.4
      f1 : nan
      mcc : nan
    removed_feature : nan
      top_feature : nan
    hyperparameter_regime : nan

[record 13 of 24]
    dataset : phiusiil
    iteration : 12
    n_features : 38
    accuracy : 0.992567
    fit_seconds : 31.7
    shap_seconds : 1597.4
    mean_tree_depth : 39.31
    total_nodes : 1738726
    window_start : 15:28:50
    window_end : 15:56:00
    cache_replay : False
    source_log : a13_resume.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : B1;B2;B2acq
    contention_evidence : phase_a.log+sampler
    sampled_cpu_percent_mean : 38.08
    sampled_cpu_percent_max : 100.0
    sampled_n_python_procs_max : 53.0
    sampled_job_tags : B2
    n_load_samples : 46.0
    contended : True
    cost_analysis_eligible : False
      remeasured_clean : False
    shap_seconds_original_contended : nan
      f1 : nan
      mcc : nan
    removed_feature : nan
      top_feature : nan
    hyperparameter_regime : nan

[record 14 of 24]
    dataset : phiusiil
    iteration : 13
    n_features : 37
    accuracy : 0.991801
    fit_seconds : 24.7
    shap_seconds : 1571.5
    mean_tree_depth : 39.632
    total_nodes : 1866634
    window_start : 15:56:00
    window_end : 16:22:37
    cache_replay : False
    source_log : a13_resume.log
    run_date : 2026-09-11
    window_reconstructed : nan
    competing_jobs : B2;B2acq
    contention_evidence : phase_a.log+sampler
    sampled_cpu_percent_mean : 19.24

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

sampled_cpu_percent_max : 100.0
sampled_n_python_procs_max : 49.0
  sampled_job_tags : B2
    n_load_samples : 76.0
      contended : True
cost_analysis_eligible : False
  remeasured_clean : False
shap_seconds_original_contended : nan
    f1 : nan
      mcc : nan
        removed_feature : nan
          top_feature : nan
            hyperparameter_regime : nan

[record 15 of 24]
  dataset : phiusiil
    iteration : 14
      n_features : 36
        accuracy : 0.989191
          fit_seconds : 33.0
            shap_seconds : 2022.1
              mean_tree_depth : 42.442
                total_nodes : 2247592
                  window_start : 16:22:37
                    window_end : 16:56:53
                      cache_replay : False
                        source_log : a13_resume.log
                          run_date : 2026-09-11
                            window_reconstructed : nan
                              competing_jobs : nan
                                contention_evidence : none
                                  sampled_cpu_percent_mean : 12.92
                                    sampled_cpu_percent_max : 100.0
                                      sampled_n_python_procs_max : 25.0
                                        sampled_job_tags : nan
                                          n_load_samples : 97.0
                                            contended : False
                                              cost_analysis_eligible : True
                                                remeasured_clean : False
shap_seconds_original_contended : nan
    f1 : nan
      mcc : nan
        removed_feature : nan
          top_feature : nan
            hyperparameter_regime : nan

[record 16 of 24]
  dataset : uci379_website_phishing
    iteration : 0
      n_features : 9
        accuracy : 0.9064039408866996
          fit_seconds : 1.77
            shap_seconds : 0.01
              mean_tree_depth : 11.693333333333332
                total_nodes : 54348
                  window_start : nan
                    window_end : nan
                      cache_replay : False
                        source_log : screening
                          run_date : nan
                            window_reconstructed : nan
                              competing_jobs : nan

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

    contention_evidence : not_assessed
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
    sampled_job_tags : nan
    n_load_samples : nan
    contended : nan
    cost_analysis_eligible : nan
    remeasured_clean : nan
    shap_seconds_original_contended : nan
    f1 : 0.9243027888446216
    mcc : 0.8056806781916287
    removed_feature : nan
    top_feature : SFH
    hyperparameter_regime : retuned_per_feature_set

```

[record 17 of 24]

```

    dataset : uci379_website_phishing__suspicious_dropped
    iteration : 0
    n_features : 9
    accuracy : 0.9202127659574468
    fit_seconds : 1.23
    shap_seconds : 0.01
    mean_tree_depth : 9.48
    total_nodes : 9690
    window_start : nan
    window_end : nan
    cache_replay : False
    source_log : screening
    run_date : nan
    window_reconstructed : nan
    competing_jobs : nan
    contention_evidence : not_assessed
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
    sampled_job_tags : nan
    n_load_samples : nan
    contended : nan
    cost_analysis_eligible : nan
    remeasured_clean : nan
    shap_seconds_original_contended : nan
    f1 : 0.9302325581395348
    mcc : 0.8375517998895313
    removed_feature : nan
    top_feature : SFH
    hyperparameter_regime : retuned_per_feature_set

```

[record 18 of 24]

```

    dataset : mendeley_hannousse
    iteration : 0
    n_features : 87
    accuracy : 0.965597667638484
    fit_seconds : 2.96
    shap_seconds : 0.01
    mean_tree_depth : 21.58
    total_nodes : 472714
    window_start : nan
    window_end : nan
    cache_replay : False
    source_log : screening
    run_date : nan

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

window_reconstructed : nan
  competing_jobs : nan
  contention_evidence : not_assessed
sampled_cpu_percent_mean : nan
sampled_cpu_percent_max : nan
sampled_n_python_procs_max : nan
  sampled_job_tags : nan
  n_load_samples : nan
    contended : nan
cost_analysis_eligible : nan
  remeasured_clean : nan
shap_seconds_original_contended : nan
  f1 : 0.9658762290341236
  mcc : 0.9313385142064948
  removed_feature : nan
  top_feature : page_rank
hyperparameter_regime : retuned_per_feature_set

```

[record 19 of 24]

```

  dataset : mendeley_tan
  iteration : 0
  n_features : 48
  accuracy : 0.9873333333333332
  fit_seconds : 1.9
  shap_seconds : 0.01
mean_tree_depth : 19.966666666666665
  total_nodes : 234594
  window_start : nan
  window_end : nan
  cache_replay : False
  source_log : screening
  run_date : nan
window_reconstructed : nan
  competing_jobs : nan
  contention_evidence : not_assessed
sampled_cpu_percent_mean : nan
sampled_cpu_percent_max : nan
sampled_n_python_procs_max : nan
  sampled_job_tags : nan
  n_load_samples : nan
    contended : nan
cost_analysis_eligible : nan
  remeasured_clean : nan
shap_seconds_original_contended : nan
  f1 : 0.9873248832555036
  mcc : 0.9746675330381922
  removed_feature : PctExtHyperlinks
  top_feature : PctExtHyperlinks
hyperparameter_regime : retuned_per_feature_set

```

[record 20 of 24]

```

  dataset : mendeley_tan
  iteration : 1
  n_features : 47
  accuracy : 0.9826666666666668
  fit_seconds : 1.85
  shap_seconds : 338.12
mean_tree_depth : 20.0
  total_nodes : 271312
  window_start : nan
  window_end : nan
  cache_replay : False

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

        source_log : screening
        run_date : nan
    window_reconstructed : nan
        competing_jobs : nan
        contention_evidence : not_assessed
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
        sampled_job_tags : nan
        n_load_samples : nan
        contended : nan
    cost_analysis_eligible : nan
        remeasured_clean : nan
    shap_seconds_original_contended : nan
        f1 : 0.9826666666666668
        mcc : 0.9653333333333334
    removed_feature : PctExtNullSelfRedirectHyperlinksRT
    top_feature : PctExtNullSelfRedirectHyperlinksRT
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 21 of 24]

```

        dataset : mendeley_tan
        iteration : 2
        n_features : 46
        accuracy : 0.9806666666666668
    fit_seconds : 2.32
    shap_seconds : 356.45
    mean_tree_depth : 20.0
        total_nodes : 310536
        window_start : nan
        window_end : nan
        cache_replay : False
        source_log : screening
        run_date : nan
    window_reconstructed : nan
        competing_jobs : nan
        contention_evidence : not_assessed
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
        sampled_job_tags : nan
        n_load_samples : nan
        contended : nan
    cost_analysis_eligible : nan
        remeasured_clean : nan
    shap_seconds_original_contended : nan
        f1 : 0.980653769179453
        mcc : 0.9613341878529912
    removed_feature : PctNullSelfRedirectHyperlinks
    top_feature : PctNullSelfRedirectHyperlinks
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 22 of 24]

```

        dataset : mendeley_tan
        iteration : 3
        n_features : 45
        accuracy : 0.9753333333333334
    fit_seconds : 2.24
    shap_seconds : 372.43
    mean_tree_depth : 20.0
        total_nodes : 343486
        window_start : nan

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

        window_end : nan
        cache_replay : False
        source_log : screening
        run_date : nan
    window_reconstructed : nan
        competing_jobs : nan
        contention_evidence : not_assessed
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
        sampled_job_tags : nan
        n_load_samples : nan
            contended : nan
    cost_analysis_eligible : nan
        remeasured_clean : nan
    shap_seconds_original_contended : nan
        f1 : 0.9752839011356046
        mcc : 0.9506742720912652
    removed_feature : FrequentDomainNameMismatch
    top_feature : FrequentDomainNameMismatch
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 23 of 24]

```

        dataset : mendeley_tan
        iteration : 4
        n_features : 44
        accuracy : 0.972
        fit_seconds : 2.22
    shap_seconds : 381.82
    mean_tree_depth : 20.0
    total_nodes : 382488
        window_start : nan
        window_end : nan
        cache_replay : False
        source_log : screening
        run_date : nan
    window_reconstructed : nan
        competing_jobs : nan
        contention_evidence : not_assessed
    sampled_cpu_percent_mean : nan
    sampled_cpu_percent_max : nan
    sampled_n_python_procs_max : nan
        sampled_job_tags : nan
        n_load_samples : nan
            contended : nan
    cost_analysis_eligible : nan
        remeasured_clean : nan
    shap_seconds_original_contended : nan
        f1 : 0.9720372836218376
        mcc : 0.9440033564623456
    removed_feature : PctExtResourceUrls
    top_feature : PctExtResourceUrls
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 24 of 24]

```

        dataset : mendeley_tan
        iteration : 5
        n_features : 43
        accuracy : 0.9646666666666668
        fit_seconds : 2.19
    shap_seconds : 414.92
    mean_tree_depth : 20.0

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```
total_nodes : 423066
window_start : nan
window_end : nan
cache_replay : False
source_log : screening
run_date : nan
window_reconstructed : nan
competing_jobs : nan
contention_evidence : not_assessed
sampled_cpu_percent_mean : nan
sampled_cpu_percent_max : nan
sampled_n_python_procs_max : nan
sampled_job_tags : nan
n_load_samples : nan
contended : nan
cost_analysis_eligible : nan
remeasured_clean : nan
shap_seconds_original_contended : nan
f1 : 0.9650165016501652
mcc : 0.9295192557785932
removed_feature : nan
top_feature : ExtMetaScriptLinkRT
hyperparameter_regime : tuned_on_full_feature_set
```

Table T13. Directional agreement per shared concept, ranked by how much the disagreement matters.

[record 1 of 9]
concept : anchor_external_ratio
rho_phiusiil : 0.0872798521005418
rho_uci : -0.7012072591666815
direction_agrees : False
abs_rho_phiusiil : 0.0872798521005418
abs_rho_uci : 0.7012072591666815
status : INVERTED
importance_weight : 0.7012072591666815
headline : True

[record 2 of 9]
concept : iframe
rho_phiusiil : 0.521331928092635
rho_uci : 0.0033935238644643
direction_agrees : False
abs_rho_phiusiil : 0.521331928092635
abs_rho_uci : 0.0033935238644643
status : NO_SIGNAL_UCI
importance_weight : 0.521331928092635
headline : True

[record 3 of 9]
concept : ssl_state
rho_phiusiil : -0.6124154229775562
rho_uci : -0.735813975062039
direction_agrees : True
abs_rho_phiusiil : 0.6124154229775562
abs_rho_uci : 0.735813975062039
status : agrees
importance_weight : 0.735813975062039
headline : False

[record 4 of 9]
concept : url_length
rho_phiusiil : -0.3539937654486287
rho_uci : -0.0533315794339325
direction_agrees : True
abs_rho_phiusiil : 0.3539937654486287
abs_rho_uci : 0.0533315794339325
status : agrees
importance_weight : 0.3539937654486287
headline : False

[record 5 of 9]
concept : subdomain_depth
rho_phiusiil : -0.1141491829083289
rho_uci : -0.3046922676521399
direction_agrees : True
abs_rho_phiusiil : 0.1141491829083289
abs_rho_uci : 0.3046922676521399
status : agrees
importance_weight : 0.3046922676521399
headline : False

[record 6 of 9]
concept : form_handler
rho_phiusiil : 0.1673089014798561
rho_uci : -0.2191900653063038
direction_agrees : False
abs_rho_phiusiil : 0.1673089014798561
abs_rho_uci : 0.2191900653063038

Table T13. Directional agreement per shared concept, ranked by how much the disagreement matters. (continued)

```

      status : INVERTED
importance_weight : 0.2191900653063038
      headline : False

[record 7 of 9]
      concept : popup
      rho_phiusiil : 0.1925460026806499
      rho_uci : -8.588679256217027e-05
direction_agrees : False
abs_rho_phiusiil : 0.1925460026806499
abs_rho_uci : 8.588679256217027e-05
      status : NO_SIGNAL_UCI
importance_weight : 0.1925460026806499
      headline : False

[record 8 of 9]
      concept : ip_in_domain
      rho_phiusiil : -0.0605489999124573
      rho_uci : -0.0941600949562038
direction_agrees : True
abs_rho_phiusiil : 0.0605489999124573
abs_rho_uci : 0.0941600949562038
      status : agrees
importance_weight : 0.0941600949562038
      headline : False

[record 9 of 9]
      concept : url_redirect
      rho_phiusiil : 0.0480143528227923
      rho_uci : 0.0201134616575737
direction_agrees : True
abs_rho_phiusiil : 0.0480143528227923
abs_rho_uci : 0.0201134616575737
      status : agrees
importance_weight : 0.0480143528227923
      headline : False

```

Table T14. Concepts retained and lost when PhiUSIL is restricted to its leak-controlled feature set.

```
[record 1 of 9]
  concept : ip_in_domain
  status : retained_after_deleaking
  reason : nan
  rho_uci : -0.0941600949562038
  rho_phiusiil : -0.0605489999124573
  polarity_status : agrees

[record 2 of 9]
  concept : url_length
  status : retained_after_deleaking
  reason : nan
  rho_uci : -0.0533315794339325
  rho_phiusiil : -0.3539937654486287
  polarity_status : agrees

[record 3 of 9]
  concept : subdomain_depth
  status : retained_after_deleaking
  reason : nan
  rho_uci : -0.3046922676521399
  rho_phiusiil : -0.1141491829083289
  polarity_status : agrees

[record 4 of 9]
  concept : iframe
  status : retained_after_deleaking
  reason : nan
  rho_uci : 0.0033935238644643
  rho_phiusiil : 0.521331928092635
  polarity_status : NO_SIGNAL_UCI

[record 5 of 9]
  concept : popup
  status : retained_after_deleaking
  reason : nan
  rho_uci : -8.588679256217027e-05
  rho_phiusiil : 0.1925460026806499
  polarity_status : NO_SIGNAL_UCI

[record 6 of 9]
  concept : url_redirect
  status : retained_after_deleaking
  reason : nan
  rho_uci : 0.0201134616575737
  rho_phiusiil : 0.0480143528227923
  polarity_status : agrees

[record 7 of 9]
  concept : form_handler
  status : retained_after_deleaking
  reason : nan
  rho_uci : -0.2191900653063038
  rho_phiusiil : 0.1673089014798561
  polarity_status : INVERTED

[record 8 of 9]
  concept : ssl_state
  status : LOST_to_deleaking
  reason : source feature(s) removed by de-leaking: ['IsHTTPS']
  rho_uci : -0.735813975062039
  rho_phiusiil : -0.6124154229775562
```

Table T14. Concepts retained and lost when PhiUSIL is restricted to its leak-controlled feature set. (continued)

polarity_status : agrees

[record 9 of 9]

concept	: anchor_external_ratio
status	: LOST_to_deleaking
reason	: source feature(s) removed by de-leaking: ['NoOfExternalRef', 'NoOfSelfRef']
rho_uci	: -0.7012072591666815
rho_phiusiil	: 0.0872798521005418
polarity_status	: INVERTED

Table T16. Runtime observations for the reproducibility section, each labelled clean or contended.

```
[record 1 of 1]
  observation : uci_full E2 explanation cost rose ~3.5x against the same data under the previous
                hyperparameters
measured_before_seconds : 447.0
measured_after_seconds : 1581.3
  ratio : 3.54
  driver : RandomForest n_estimators 100 -> 500
  driver_exact : True
  driver_detail : uci_full received its own hyperparameter search in A4 (justified: it carries 11,055 rows
                  against uci_dedup's 5,849). That search selected 500 trees where the earlier uci parameters
                  used 100 - a 5x increase in the forest the explainer must walk.
  mechanism : Interventional TreeSHAP cost scales with the total node count of the model being explained,
              so explanation cost tracks model complexity regardless of WHY the complexity changed.
  significance : An INDEPENDENT confirmation of the B5b mechanism through a different driver. In A1.3 node
                 count rose because features were removed and the boundary hardened; here it rose because a
                 hyperparameter search chose a larger forest. Same relationship, unrelated cause - which is
                 what makes explanation cost a general proxy for model complexity rather than an artifact of
                 the removal procedure.
before_timing_clean : False
after_timing_clean : False
  contention_note : NEITHER timing is uncontended: the 447s baseline ran alongside A1.3, and the 1581s figure
                  ran alongside table-building commands (load sampler shows job tags A4A5+B2 across
                  18:38-19:05). The 5x tree count is exact and structural; the ~3.5x ratio is approximate and
                  should be quoted as 'roughly 3.5x' or re-measured on an idle machine before being stated
                  precisely.
  n_explained : 5527
    source : A5 uci_full seed 42/43, half A
```